

Statistical comparisons and full metrics

Accuracy: paired differences between LV and each external competitor. Units are F1 percentage points; positive differences favour LV.

Design / competitor	Mean difference	Lower bound*	Bootstrap lower	Holm p	Supported
Random / Shepherd	n/a	n/a	n/a	n/a	n/a
Random / Starcode sphere	+0.04468	+0.03322	+0.03372	4.36e-14	Yes
Random / Starcode MP	+0.58456	+0.55942	+0.56049	4.48e-54	Yes
Random / Bartender	+0.14975	+0.13852	+0.13915	2.62e-40	Yes
Anchored / Shepherd	n/a	n/a	n/a	n/a	n/a
Anchored / Starcode sphere	+0.04295	+0.03084	+0.03138	1e-12	Yes
Anchored / Starcode MP	+0.79053	+0.75721	+0.75913	1.63e-54	Yes
Anchored / Bartender	+0.56509	+0.54439	+0.54550	5.28e-58	Yes

*Simultaneous one-sided paired-t lower bounds use Bonferroni alpha/8 (familywise alpha 0.05). The paired bootstrap resamples whole libraries 100,000 times. A superiority label requires both lower bounds above zero. Incomplete contrasts are n/a; all eight positions remain in the correction. The failure-reporting addendum was introduced during execution. Milo is excluded from these tests.

The sample size was fixed at 60 per design after a four-library development variance pilot. The resource cap was binding: the planned 0.01 percentage-point difference is not detectable with 90% power in every contrast. These uncertainty estimates rely on the stated statistical assumptions; no sample-size extension followed final results.

Read assignments and abundance

Design / method	Correct reads %	Wrong reads	Unassigned	Abundance TV
Random / barbac Hamming	99.997077	29.23	0.00	0.00002923
Random / barbac LV + Poisson	99.997083	29.17	0.00	0.00002917
Random / Shepherd	n/a	n/a	n/a	n/a
Random / Starcode sphere	99.982520	174.78	0.00	0.00017480
Random / Starcode MP	99.983934	160.65	0.00	0.00016066
Random / Bartender	99.930972	690.32	0.00	0.00068973
Anchored / barbac Hamming	99.993137	68.62	0.00	0.00006863
Anchored / barbac LV + Poisson	99.994257	57.42	0.00	0.00005743
Anchored / Shepherd	n/a	n/a	n/a	n/a
Anchored / Starcode sphere	99.986583	134.17	0.00	0.00013417
Anchored / Starcode MP	99.979174	208.25	0.00	0.00020826
Anchored / Bartender	99.906649	933.50	0.00	0.00093298
Milo / barbac Hamming	99.999088	228.00	0.00	0.00000908
Milo / barbac LV + Poisson	99.999476	131.00	0.00	0.00000520
Milo / Shepherd	99.999132	129.00	88.00	0.00000864
Milo / Starcode sphere	99.847880	38024.00	0.00	0.00152108
Milo / Starcode MP	99.981789	4552.00	0.00	0.00018207
Milo / Bartender	99.935342	16162.00	0.00	0.00063614

Simulation entries are means over all 60 independent libraries; Milo entries are one fixed reference. Wrong reads exclude unassigned reads, which are counted separately. Read accuracy counts both as failures. Abundance TV is half the normalized absolute count difference, including unassigned reads.

The archived Milo truth counts and origin labels differ at four parents (six reads in total absolute count difference); total reads and identities reconcile. This preserved source discrepancy limits interpretation of very small abundance differences.

Timing, simulation limits and reproducibility

Design / competitor	LV / competitor time	95% interval
Random / Shepherd	n/a	n/a
Random / Starcode sphere	0.545	0.532 to 0.558
Random / Starcode MP	0.559	0.546 to 0.573
Random / Bartender	0.940	0.921 to 0.959
Anchored / Shepherd	n/a	n/a
Anchored / Starcode sphere	0.548	0.533 to 0.563
Anchored / Starcode MP	0.553	0.518 to 0.590
Anchored / Bartender	0.880	0.858 to 0.903

A time ratio below one favours LV. Ratios are geometric means of within-library workflow ratios. Intervals use paired log times and are secondary descriptive 95% intervals; they are not the multiplicity-adjusted primary accuracy tests.

Timing repair: the Mac slept and partly woke during one block. The 28 successful cells whose receipt intervals overlapped that power-state window were selected before retiming, independently of their results. Their repaired observations replace the original timing without selecting the faster value. Clustering outputs must be byte-identical; accuracy is unchanged. The original single-invocation plan and all measurements remain preserved. See `TIMING_REPAIR.md` and `timing_repair_protocol.json`.

Simulation boundary

Seed	Design	Boundary reads	Total reads
2026091204	anchored_mixed	1	999,064
2026091205	anchored_mixed	11	999,555
20260912303	anchored_mixed	126	1,002,454
20260912306	anchored_mixed	2	1,000,767
20260912308	random_mixed	2	1,000,828
20260912314	anchored_mixed	10	999,169
20260912322	anchored_mixed	2	1,000,041
20260912323	anchored_mixed	1	999,129
20260912328	anchored_mixed	1	1,000,828
20260912337	anchored_mixed	5	999,524
20260912338	anchored_mixed	1	997,969
20260912342	anchored_mixed	2	999,303
20260912346	anchored_mixed	72	999,192
20260912351	anchored_mixed	9	999,529
20260912352	anchored_mixed	1	1,000,008
20260912355	anchored_mixed	3	1,001,108

Previous error-rate and genuine-neighbour challenges

The earlier 0x/1x/3x/10x pilot retains all 24 planned cells, including seven unsupported cases under the previous strict simulator. The rejected paired-indel candidate corrected some errors but lost 31 true identities in six deliberately enriched genuine-neighbour controls. That candidate is excluded here; released v14 preserved those identities. These controls characterize failure modes, not real-world prevalence.

Reproducibility

Code/protocol commit: `fe05e8e873da0980ed1f100ff79217de791f7f77`. Frozen at: 2026-09-12T20:19:42.239953+00:00. Full parameters, sample-size calculations, seeds, environment, binary fingerprints, commands, individual metrics and output hashes accompany this table. Existing main, previous benchmark outputs and earlier candidate branches are preserved.

Sources: Johnson, Venkataram and Kryazhimskiy (2023), Best Practices in Designing, Sequencing, and Identifying Random DNA Barcodes (PMC10276077); archived analysis, Zenodo 7411747. Simulation-study design guidance: Morris, White and Crowther (2019), doi:10.1002/sim.8086. Our updated-coordinate and finite-support simulation is an explicitly described adaptation.

Retained tool failures

These registered calls did not produce a valid clustering output. Inputs, commands and logs remain preserved. No failing call was rerun with altered settings and no failed score was replaced by zero. A success-only average is conditional on the method completing and does not establish full-design accuracy or runtime.

Design / method	Seed	Failure
random_mixed / Shepherd	20260912343	Automatic error-rate estimate rejected
random_mixed / Shepherd	20260912301	Automatic error-rate estimate rejected
anchored_mixed / Shepherd	20260912302	Automatic error-rate estimate rejected
random_mixed / Shepherd	20260912302	Automatic error-rate estimate rejected
random_mixed / Shepherd	20260912311	Automatic error-rate estimate rejected
random_mixed / Shepherd	20260912326	Automatic error-rate estimate rejected
anchored_mixed / Shepherd	20260912311	Automatic error-rate estimate rejected
anchored_mixed / Shepherd	20260912326	Automatic error-rate estimate rejected
anchored_mixed / Shepherd	20260912343	Automatic error-rate estimate rejected
anchored_mixed / Shepherd	20260912301	Automatic error-rate estimate rejected

Successful-only summaries for incomplete rows

Design / method	Succeeded	FN	FP	F1 %	Time s
random_mixed / Shepherd	55/60	124.15	64.84	99.05236	3.36
anchored_mixed / Shepherd	55/60	126.87	24.53	99.23909	9.92

FN, FP and F1 above are means over successful calls only; time is their median. They use fewer libraries than the main table and must not be compared as if they represented the same complete population. Full per-input results retain every failure.

Failure handling: FAILURE_REPORTING.md and analyze_available.py were added after the first failure. The original frozen analyze.py is retained. The complete-pair calculations, bootstrap seeds and correction family are unchanged; unavailable contrasts have no superiority conclusion. Pawel et al. (2025), Handling Missingness, Failures, and Non-Convergence in Simulation Studies, arXiv:2409.18527v3, discusses the consequences of excluding or replacing failed outputs.

Comparison on the same successful inputs

Random, the same 55 successful inputs: LV mean F1 99.29302%; Shepherd 99.05236%. These matched-subset means exclude failed inputs and are descriptive only; they do not replace the unavailable full-design test.

Anchored, the same 55 successful inputs: LV mean F1 99.25247%; Shepherd 99.23909%. These matched-subset means exclude failed inputs and are descriptive only; they do not replace the unavailable full-design test.